Midpoint of a segment

1

a $M\,(5,-2)$

b $M\,(10,6)$

2

a $M\,(5,7)$

b $M\left(\dfrac{-11}{2},-8\right)$

c $M\,(-3,2)$

d $M\,(-3,-2)$

e $M\,(-2,1)$

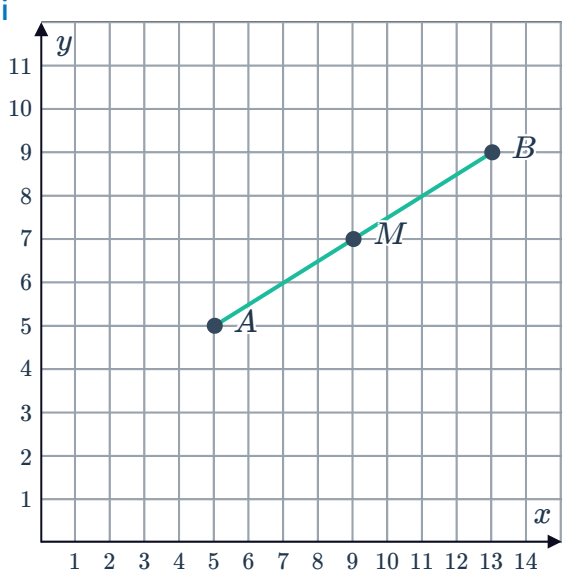
f $M\,(2,-4)$

3

a

i $M\,(9,7)$

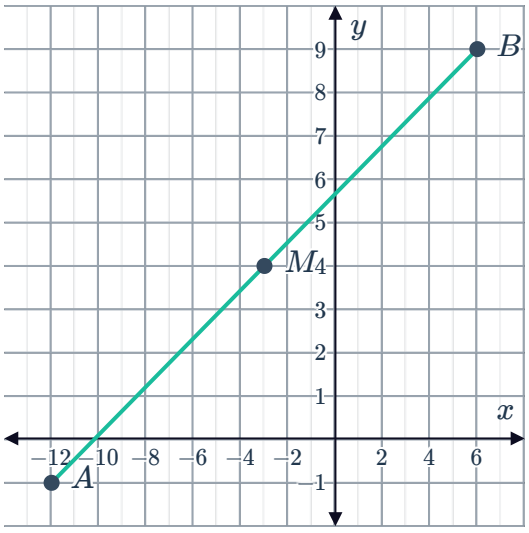
ii



b

i $M\,(-3,4)$

ii

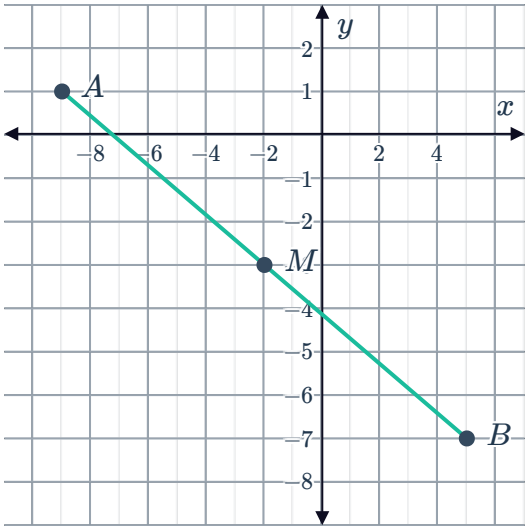


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c

i $M\,(-2,-3)$

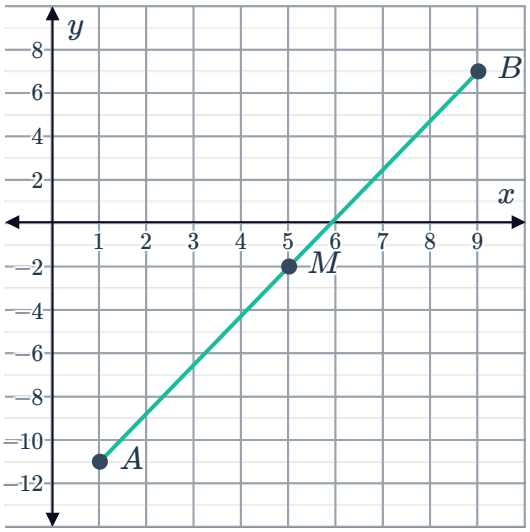
ii



d

i $M\,(5,-2)$

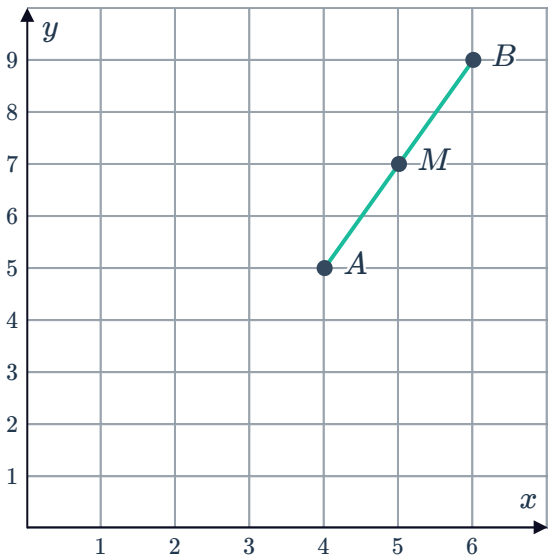
ii



e

i $M\,(5,7)$

ii

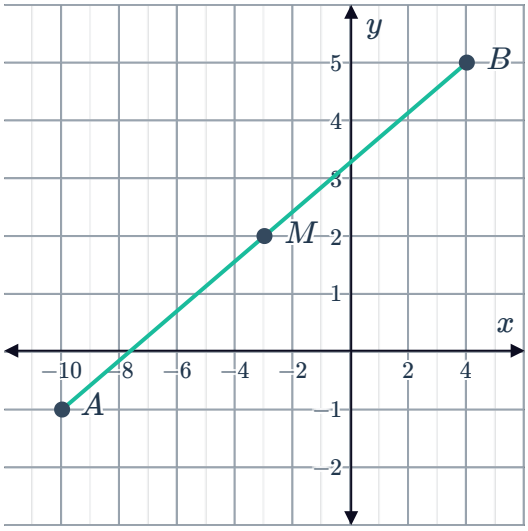


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f

i $M\,(-3, 2)$

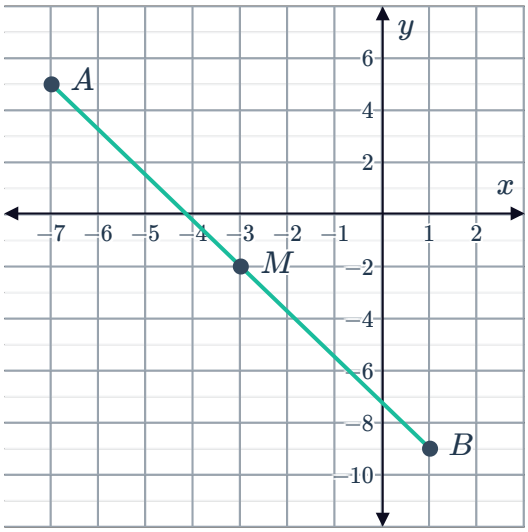
ii



g

i $M\,(-3, -2)$

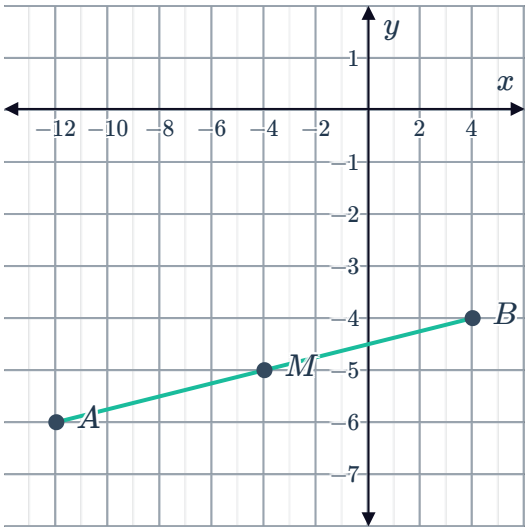
ii



h

i $M\,(-4, -5)$

ii



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4

a $M(8,6)$ b $M\left(\frac{7}{2},\frac{1}{2}\right)$ c $M(-5,9)$ d $M(6,-9)$

e $M(-9,-4)$ f $M(-5,2)$ g $M(2,-6)$ h $M(-4,-7)$

i $M\left(-\frac{3}{4},\frac{3}{4}\right)$ j $M\left(3,\frac{5}{2}\right)$

5 $(4m,3n)$

6 $\left(m-3,\frac{31n}{12}\right)$

7

a $p=-\frac{1}{4}$ b $q=0$

8

a $M(3,7)$ b $k=-11$

9 $Q(-6,1), R(-8,4), S(-10,7)$

End points of a segment

10

a $A(5,6)$ b $A(6,-5)$ c $A(5,6)$ d $A(5,-6)$

e $A(-5,6)$ f $A(4,-3)$ g $A(-4,-5)$ h $A(6,22)$

i $A(-3,3)$ j $A(-7,-4)$

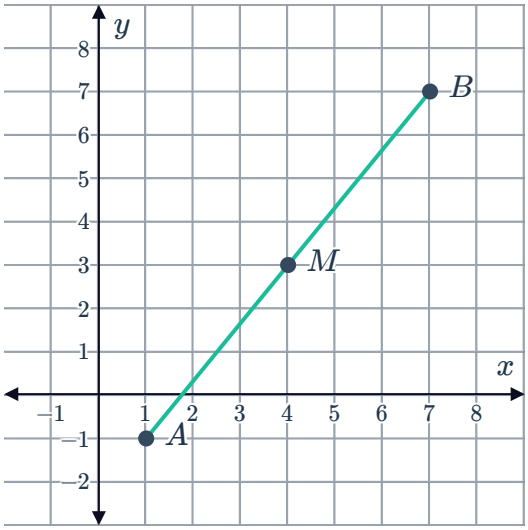
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11

a

i $B\,(7, 7)$

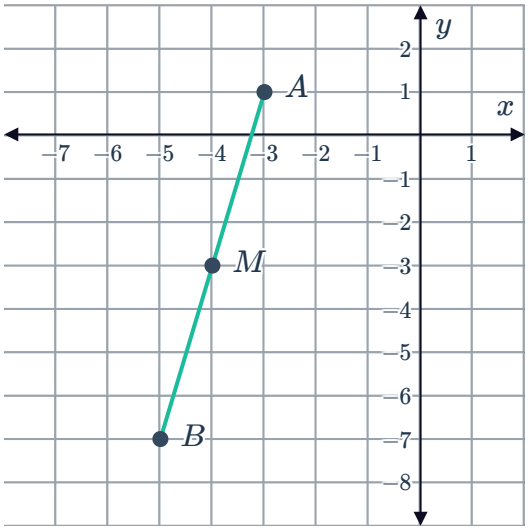
ii



b

i $B\,(-5, -7)$

ii



12 $B\,(8, -1)$

13

a Their diagonals bisect one another.

b **i** $\left(-\frac{17}{2}, 3\right)$ **ii** $\left(-\frac{17}{2}, 3\right)$

c Yes

14

a $T\,(8, 8)$

b $S\,(6, 1)$

c Parallelogram, because the diagonals bisect each other.

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Applications

15 29 million

16 63 years

17 $\left(\frac{69}{2}^\circ\text{N}, 103^\circ\text{E}\right)$

18

a $207\,203$

b $268\,291$

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